

Data Encryption and Decryption

Encryption:

- Encryption is an algorithm that converts the plain text into cipher text by using secret key.
- Encryption algorithm takes plain text which is in readable information and converts them into cipher text (unreadable information) with the help of the shared secret key.

Decryption:

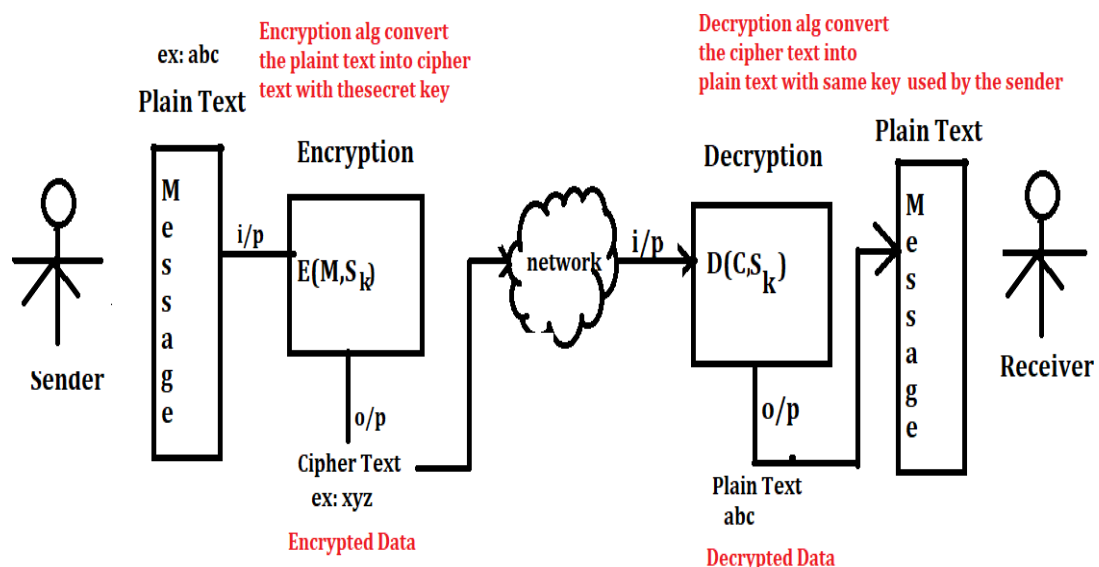
- Decryption is also a algorithm that takes cipher text and convert it into plain text using key.
- Decryption works reverse of encryption algorithm. I takes cipher text and return plain text
- Encryption performed by sender where decryption is performed by receiver.

Types of Encryption:

- ❖ Symmetric Key Encryption (AES)
- ❖ Asymmetric Key Encryption (RSA)

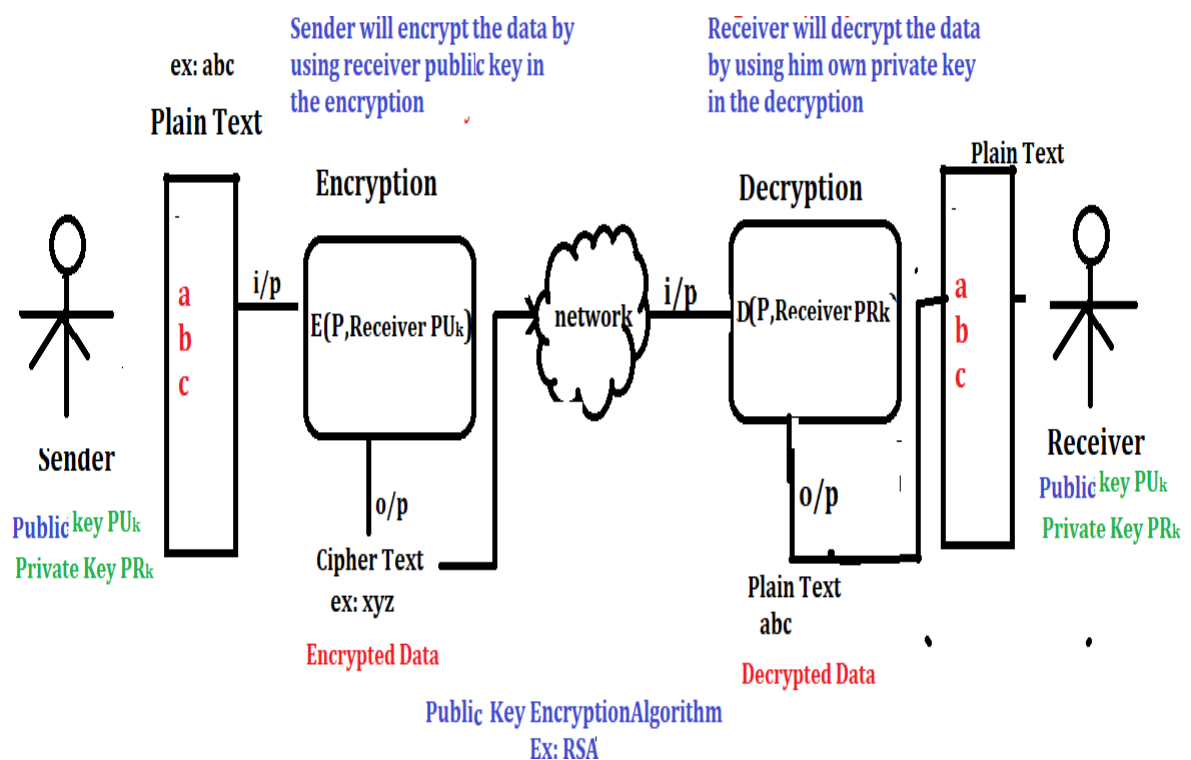
1. Symmetric Key Encryption:

- Symmetric key encryption is a type of encryption in which both sender and receiver shared same secret key in order to encrypt and decrypt the message
- Advanced Encryption Standard (AES) is the best and fastest algorithm comes under symmetric key encryption.



2. Asymmetric Key Encryption:

- Asymmetric Key Encryption algorithm is also known as Public key encryption algorithm in which both sender and receiver will use two different keys in order to encrypt and decrypt the message.
- In the public key encryption, Sender will encrypt the plain text into cipher text by using receiver public key where as receiver will decrypt the cipher text by using him/his own private key.
- Hence in public key encryption we are using two different keys such as public key and private key
- Public key is a key which is every one known where as private key is a key that kept it secret.
- RSA is a best example algorithm of public key encryption.



1. Confidentiality in Encryption

2. Authentication in Encryption

- ➔ Confidentiality means transmit the data from sender to receiver in secure channel. Confidentiality helps you to protect the data from unauthorized people.
- ➔ Authentication/Integrity help to check whether the received data is came from alleged source or not and also helps to whether the data is modified by someone or not .

Digital Signature:

Digital Signature is a mechanism used to verify whether the received data came from the **correct/authentic sender** and whether the data has been **modified during transmission**.

A digital signature uses a **hash function and asymmetric cryptography** to provide:

- **Authentication** – verifies the identity of the sender.
- **Data Integrity** – ensures that the data has not been modified.
- **Non-repudiation** – provides evidence that the sender signed the data.

How Digital Signature Works

1. The **sender prepares the original message**.
2. The sender applies a **hash function** to the message.

Message → Hash Function → Hash Value

3. The sender encrypts the hash value using the **sender's private key**. This encrypted hash is called the **Digital Signature**.

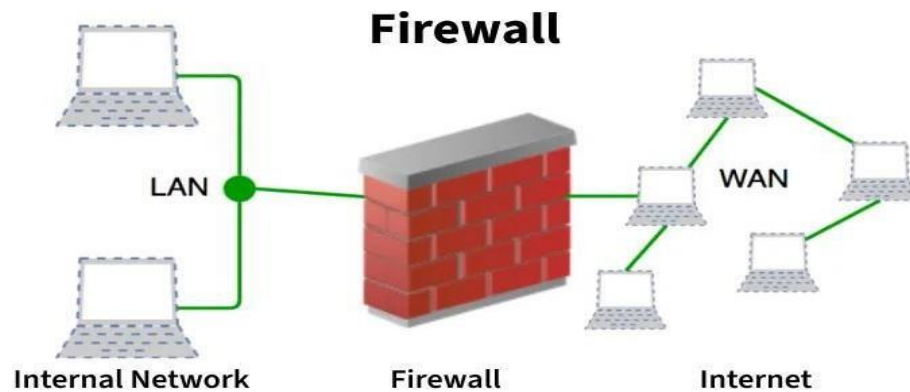
Hash Value + Sender's Private Key → Digital Signature

4. The sender sends the **original message + digital signature** to the receiver.
5. The receiver uses the **sender's public key** to verify/decrypt the digital signature and obtain the original hash value.
6. The receiver independently applies the **same hash function** to the received message to generate a new hash value.
7. The receiver compares the two hash values:
 - **Hash values match** → message is authentic and has not been modified.
 - **Hash values do not match** → message may have been modified or the signature is not valid.

Therefore, digital signatures help achieve **authentication, data integrity, and non-repudiation**.

Firewall:

- ➔ Firewall is a software or hardware that monitors and control the incoming and outgoing network traffic based on the predefined security rules.
- ➔ Firewall is a network security device that acts as a wall in between Private network and public network in order to prevent the un-authorized access.



➔ It examines the data packets while entering and leaving the traffic and also allows the safe data packets and block the suspicious or unsafe data packets from the traffic.

➔ In simply, Firewall protects the organization data when we are connecting the internet.

Internal Network ←Firewall→ Internet (external network)

Protocols in Networking

A **network protocol** is a set of **rules and standards** that devices follow to communicate with each other over a network.

Protocol	Full Form	Main Purpose
HTTP	HyperText Transfer Protocol	Transfers web pages
HTTPS	HyperText Transfer Protocol Secure	Secure web communication
FTP	File Transfer Protocol	Transfers files
SMTP	Simple Mail Transfer Protocol	Sends emails
POP3	Post Office Protocol 3	Downloads emails
IMAP	Internet Message Access Protocol	Accesses/manages emails
DNS	Domain Name System	Converts domain names to IP addresses
DHCP	Dynamic Host Configuration Protocol	Automatically assigns IP addresses
TCP	Transmission Control Protocol	Reliable data delivery
UDP	User Datagram Protocol	Fast, connectionless data delivery
IP	Internet Protocol	Provides addressing and routing
ARP	Address Resolution Protocol	Finds MAC address from IP address
ICMP	Internet Control Message Protocol	Network error/reporting and diagnostic messages
SSH	Secure Shell	Secure remote access
Telnet	Telecommunication Network	Remote access, but not secure

What is Port?

- ➔ In networking port is number that describes different services or application running on the same IP address in to the computer or mobile or any device
- ➔ There are three types of port numbers.

- ❖ Well Known Port Numbers

- Port: 80 → Web server (HTTP)
- 443 → HTTPS.
- 22 → SSH.

- ❖ Registered Port Numbers.

- 1433 → Microsoft SQL Server
- 3306 → MySQL Database
- 8080 → Alternative HTTP service

- ❖ **Random or Ephemeral Port Number:**

When your device connects to the server it needs a source port to identify the connection. Server knows only fixed/well known ports. So it picks a temporary or random port numbers called as ephemeral port start from 49152 to 65535

- **IP Address** → identifies the device in the internet (like a house address).
- **MAC Address** → identifies the device in the Ethernet/LAN network card (like a unique ID of the house).
- **Port** → identifies the specific service or application inside the device (like a room number in the house).

Example:

- Your computer's IP: 192.168.1.10 (your house).
- MAC: 08:00:27:4E:66:A1 (the unique lock/key of your house).
- Port: 80 → Web server (HTTP), 22 → SSH, 443 → HTTPS.

Let's see how data can be transformed in the below diagram:

Application (Browser) GET / HTTP/1.1



Transport Layer (Ports) SrcPort: 54321 → DstPort:80



Network Layer (IP) SrcIP: 192.168.1.10 → DstIP: 93.184.216.34



Data Link Layer (MAC)

SrcMAC: 08:00:27:4E:66:A1 → DstMAC: 00:11:22:33:44:55



Physical Layer (Bits on wire)